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Type3 Asset Administration Shell

A concept and implementation of a proactive Asset Administration Shell (AAS) with Multi-Agent-System (MAS) integration using Belief-Desire-Intention (BDI) approach

by

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Abstract

This master thesis presents the concept and implementation of a Type3 Asset Adminstartion Shell (AAS) with integration into a Multi-Agent System (MAS) using the Belief-Desire-Intention (BDI) approach. The proposed proactive AAS framework aims to enhance the interoperability and autonomy of industrial assets by leveraging agent-based technologies. The thesis details the design principles, architecture, and implementation strategies employed in developing the AAS, along with a comprehensive evaluation of its performance in a simulated industrial environment. The results demonstrate the effectiveness of the BDI-based MAS integration in enabling dynamic decision-making and adaptive behavior in response to changing operational conditions.

List of Acronyms

AAS **Asset Adminstartion Shell**

RAMI 4.0 **Reference Architectural Model Industrie 4.0**

MAS **Multi Agent system**

JADE **JAVA Agent DEvelopment Framework**

JADEX **JADE eXtension**

SPADE **Smart Python Agent Development Environment**

IIRA **Industrial Internet Reference Architecture**

CPS **Cyber Physical Systems**

I4.0 **Industry 4.0**

IDTA **Industrial Digital Twin Association**

ACL **agent communication language**

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1 Introduction

About this chapter

This chapter introduces the fundamental concepts and motivations behind the development of Type3 Asset Administration Shells (AAS) and their integration with Multi-Agent Systems (MAS) using the Belief-Desire-Intention (BDI) approach. It outlines the significance of AAS in the context of Industry 4.0, highlighting their role in enhancing asset management and interoperability. We also go over the other chapters of this thesis to give a brief overview of the content covered.

2 Literature review

About this chapter

This chapter provides a comprehensive review of the existing literature and foundational concepts relevant to the development of Type3 Asset Administration Shells (AAS) and their integration with Multi-Agent Systems (MAS) using the Belief-Desire-Intention (BDI) approach. It covers the basic principles of AAS, different types of AAS with a focus on Type3, and explores the architecture and functionalities of MAS, particularly in industrial applications. The chapter also delves into the BDI model, highlighting its significance in enabling autonomous decision-making within agent-based systems.

2.1 Asset Administration Shells(AAS)

About AAS

In this section , we explore the fundamental concepts of Asset Administration Shells (AAS), which serve as digital representations of physical assets in the context of Industry 4.0. We discuss the structure and components of AAS, including submodels and data types, and provide an overview of the different types of AAS, with a particular emphasis on Type3 AAS. This section aims to establish a solid understanding of AAS as a foundation for their integration with Multi-Agent Systems (MAS) in subsequent sections.

2.1.1 Basic concept of AAS

About AAS Concept

In this subsection, we delve into the basic concept of Asset Administration Shells (AAS), which are pivotal in the digital transformation of industrial assets. We discuss the role of AAS in providing a standardized digital representation of physical assets, enabling interoperability and communication within Industry 4.0 ecosystems. The subsection outlines the key features and functionalities of AAS, highlighting their significance in facilitating asset management, data exchange, and integration with other digital systems.

2.1.2 Submodels and data types

About Submodels and Data Types

This subsection focuses on the structure of Asset Administration Shells (AAS), specifically the concept of submodels and data types. We explore how submodels serve as modular components within the AAS, encapsulating specific functionalities and data related to the asset. The subsection also discusses various data types used in AAS, emphasizing their role in ensuring accurate representation and interoperability of asset information. This foundational understanding is crucial for comprehending the complexities of AAS and their integration with Multi-Agent Systems (MAS).

2.1.3 Types of AAS

About Types of AAS

In this subsection, we provide an overview of the different types of Asset Administration Shells (AAS) as defined in the literature. We categorize AAS into various types based on their functionalities, capabilities, and levels of integration with physical assets. The subsection highlights the distinctions between these types, with a particular focus on Type3 AAS, which is characterized by its advanced features and proactive capabilities. This classification serves as a basis for understanding the specific requirements and design considerations for implementing Type3 AAS in industrial applications.

2.1.4 AAS Type3

About AAS Type3

This subsection delves into the specifics of Type3 Asset Administration Shells (AAS), which represent an advanced category of AAS with enhanced functionalities. We discuss the defining characteristics of Type3 AAS, including their ability to support proactive behaviors, complex data management, and integration with intelligent systems. The subsection also explores the design principles and architectural considerations unique to Type3 AAS, highlighting their role in facilitating sophisticated asset management and interoperability within Industry 4.0 environments.

2.2 Multi Agent System (MAS)

About MAS

In this section, we explore the concept of Multi-Agent Systems (MAS) and their relevance in industrial applications. We discuss the fundamental principles of MAS, including agent autonomy, communication, and cooperation. The section also highlights various frameworks and architectures used in implementing MAS, emphasizing their significance in enhancing the capabilities of Asset Administration Shells (AAS) through intelligent agent-based interactions. This foundational understanding of MAS sets the stage for examining its integration with AAS in subsequent sections.

2.3 BDI (Belief-Desire-Intention) Approach

About BDI Approach

This section delves into the Belief-Desire-Intention (BDI) approach, a prominent model for designing intelligent agents within Multi-Agent Systems (MAS). We explore the core components of the BDI model, including beliefs, desires, and intentions, and discuss how these elements contribute to autonomous decision-making and goal-oriented behavior in agents. The section also highlights the advantages of using the BDI approach in industrial applications, particularly in enhancing the functionality and adaptability of Asset Administration Shells (AAS) through intelligent agent integration.

3 State of the art

About this chapter

This chapter provides a comprehensive review of the current state of the art in the fields of Asset Administration Shells (AAS), Multi-Agent Systems (MAS), and their integration. It discusses existing frameworks, methodologies, and technologies relevant to the development of Type3 AAS with BDI agent integration. The chapter identifies gaps in the current research landscape, setting the stage for the proposed contributions of this thesis.

3.1 Current frameworks for Asset Administration Shells

About Current Frameworks for AAS

This section reviews the existing frameworks and standards for Asset Administration Shells (AAS). It discusses various implementations, their features, and how they address the challenges of interoperability and asset management in Industry 4.0. The section also highlights the limitations of current frameworks, particularly in relation to Type3 AAS and their integration with intelligent systems.

3.2 Current frameworks for Muti Agent Systems and BDI Agents

About Current Frameworks for MAS and BDI Agents

This section reviews the existing frameworks and methodologies for Multi-Agent Systems (MAS) and Belief-Desire-Intention (BDI) agents. It covers various agent architectures, communication protocols, and decision-making models used in industrial applications. The section also discusses the challenges and limitations of current MAS and BDI frameworks, especially in the context of their integration with Asset Administration Shells (AAS).

3.3 Existing integrations of AAS with MAS

About Existing Integrations of AAS with MAS

This section explores the current state of integration between Asset Administration Shells (AAS) and Multi-Agent Systems (MAS). It reviews existing approaches, case studies, and implementations that combine AAS with MAS to enhance asset management and automation. The section identifies the strengths and weaknesses of these integrations, providing insights into the potential for incorporating BDI agents within Type3 AAS frameworks.

3.4 Gaps in the current state of the art

About Gaps in the Current State of the Art

This section identifies the gaps and challenges in the current research and development landscape related to Asset Administration Shells (AAS), Multi-Agent Systems (MAS), and their integration. It highlights areas where existing frameworks fall short, particularly in supporting the advanced functionalities of Type3 AAS and the incorporation of BDI agents. The section sets the stage for the proposed contributions of this thesis by outlining the unmet needs and opportunities for innovation. foucsing on the lack of integrating the AAS Operation ”methods and functions” with MAS frameworks using BDI agents.

4 System requierments

Note

This chapter outlines the system requirements for the proposed Type3 Asset Administration Shell (AAS) integrated with a Multi-Agent System (MAS) using the Belief-Desire-Intention (BDI) approach. It details both functional and non-functional requirements, providing a comprehensive overview of the necessary specifications to ensure successful implementation and operation of the system.

4.1 Functional requirements

About Functional Requirements

This section outlines the functional requirements for the Type3 Asset Administration Shell (AAS) integrated with a Multi-Agent System (MAS) using the Belief-Desire-Intention (BDI) approach. It specifies the essential functionalities that the system must possess to effectively manage and represent industrial assets, facilitate communication and cooperation among agents, and support autonomous decision-making processes. The requirements detailed in this section serve as a foundation for the design and implementation of the proposed system.

4.2 Non-functional requirements

About Non-Functional Requirements

This section delineates the non-functional requirements for the Type3 Asset Administration Shell (AAS) integrated with a Multi-Agent System (MAS) using the Belief-Desire-Intention (BDI) approach. It addresses critical aspects such as performance, scalability, security, and usability that are essential for the effective operation and adoption of the system in industrial environments. The non-functional requirements outlined in this section ensure that the system not only meets its functional objectives but also adheres to quality standards necessary for real-world applications.

4.3 Usecase

About Usecase

This section presents a detailed use case scenario for the Type3 Asset Administration Shell (AAS) integrated with a Multi-Agent System (MAS) using the Belief-Desire-Intention (BDI) approach. It illustrates how the system operates in a practical industrial context, highlighting the interactions between the AAS and BDI agents to achieve specific objectives. The use case serves to demonstrate the application of the functional and non-functional requirements outlined in previous sections, providing a concrete example of the system's capabilities and benefits.

5 Concept

About this chapter

This chapter presents the core concepts underlying the proposed Type3 Asset Administration Shell (AAS) framework and its integration with Multi-Agent Systems (MAS) using the Belief-Desire-Intention (BDI) model. It details the architectural design, key components, and interaction mechanisms that facilitate proactive behavior in industrial assets. The chapter also discusses the theoretical foundations and practical considerations involved in implementing the AAS-MAS integration.

5.1 System overview

About System Overview

This section provides an overview of the proposed Type3 Asset Administration Shell (AAS) integrated with a Multi-Agent System (MAS) using the Belief-Desire-Intention (BDI) approach. It outlines the high-level architecture, key components, and their interactions within the system. The section aims to give readers a comprehensive understanding of how the AAS and MAS work together to achieve enhanced asset management and autonomous decision-making in industrial environments.

5.2 AAS Concept

About AAS Concept

This section delves into the concept of Asset Administration Shells (AAS) as digital representations of physical assets in Industry 4.0. It discusses the structure, components, and functionalities of AAS, with a focus on Type3 AAS and its advanced capabilities. The section also explores how AAS facilitates interoperability, data exchange, and integration with intelligent systems, setting the stage for its combination with Multi-Agent Systems (MAS).

5.3 AAS-MAS Integration Concept

About AAS-MAS Integration Concept

This section explores the concept of integrating Asset Administration Shells (AAS) with Multi-Agent Systems (MAS) using the Belief-Desire-Intention (BDI) approach. It discusses the architectural design, communication mechanisms, and interaction protocols that enable seamless collaboration between AAS and BDI agents. The section highlights the benefits of this integration in enhancing asset management, autonomy, and decision-making capabilities within industrial environments.

5.4 BDI Agent Concept

About BDI Agent Concept

This section delves into the concept of Belief-Desire-Intention (BDI) agents and their role within Multi-Agent Systems (MAS). It discusses the core components of the BDI model, including beliefs, desires, and intentions, and how these elements contribute to autonomous decision-making and goal-oriented behavior in agents. The section also explores the advantages of using BDI agents in industrial applications, particularly in enhancing the functionality and adaptability of Asset Administration Shells (AAS) through intelligent agent integration.

5.5 Proactive AAS Concept

About Proactive AAS Concept

This section introduces the concept of proactive Asset Administration Shells (AAS) and their capabilities in anticipating and responding to dynamic industrial environments. It discusses the design principles, architectural considerations, and interaction mechanisms that enable AAS to exhibit proactive behavior through integration with Multi-Agent Systems (MAS) using the Belief-Desire-Intention (BDI) approach. The section highlights the benefits of proactive AAS in enhancing asset management, autonomy, and decision-making processes.

6 Implementation

About this chapter

This chapter details the implementation of the Type3 Asset Administration Shell (AAS) integrated with a Multi-Agent System (MAS) using the Belief-Desire-Intention (BDI) approach. It covers the technical aspects of the AAS development, including the software architecture, key components, and the integration process with the MAS. The chapter also discusses the challenges encountered during implementation and the solutions adopted to address them.

6.1 AAS implementation and repository design pattern

About AAS Implementation and Repository Design Pattern

This section describes the implementation of the Type3 Asset Administration Shell (AAS) and the application of the repository design pattern. It outlines the software architecture, key components, and data management strategies employed in the AAS development. The section also discusses how the repository design pattern facilitates efficient data access, storage, and retrieval within the AAS framework, enhancing its scalability and maintainability.

6.2 Resource Agents implementation

About Resource Agents Implementation

This section details the implementation of Resource Agents within the Multi-Agent System (MAS) integrated with the Type3 Asset Administration Shell (AAS). It discusses the design principles, architectural considerations, and interaction mechanisms that enable Resource Agents to effectively manage and represent physical assets. The section also highlights the role of Resource Agents in facilitating communication and cooperation among agents, as well as their contribution to the overall functionality and autonomy of the integrated AAS-MAS framework.

6.3 Production Agents implementation

About Production Agents Implementation

This section outlines the implementation of Production Agents within the Multi-Agent System (MAS) integrated with the Type3 Asset Administration Shell (AAS). It discusses the design principles, architectural considerations, and interaction mechanisms that enable Production Agents to effectively manage and represent product-related information. The section also highlights the role of Production Agents in facilitating communication and cooperation among agents, as well as their contribution to the overall functionality and autonomy of the integrated AAS-MAS framework.

6.4 MAS communication over XMPP protocols in Asynchronous way

About MAS Communication over XMPP Protocols in Asynchronous Way

This section discusses the implementation of communication mechanisms within the Multi-Agent System (MAS) integrated with the Type3 Asset Administration Shell (AAS) using XMPP protocols in an asynchronous manner . It outlines the architectural design, communication protocols, and interaction patterns that facilitate efficient and reliable message exchange among agents. The section also highlights the benefits of using XMPP for asynchronous communication, including scalability, flexibility, and real-time data transfer, which enhance the overall performance and responsiveness of the integrated AAS-MAS framework.

7 Evaluation

About this chapter

This chapter presents a comprehensive evaluation of the Type3 Asset Administration Shell (AAS) integrated with a Multi-Agent System (MAS) using the Belief-Desire-Int ention (BDI) approach. It details the methodologies employed to assess the performance, scalability, and effectiveness of the proposed framework in various industrial scenarios. The chapter also discusses the results obtained from simulations and real-world implementations, highlighting the strengths and potential areas for improvement in the AAS-MAS integration.

8 Outlook

About this chapter

This chapter provides an outlook on the future directions and potential advancements in the development and application of Type3 Asset Administration Shells (AAS) integrated with Multi-Agent Systems (MAS) using the Belief-Desire-Intention (BDI) approach. It discusses emerging trends, challenges, and opportunities in the field, as well as suggestions for further research and development to enhance the capabilities and adoption of AAS-MAS frameworks in industrial environments.

References